MSB7216: Deep Learning for Health Data

Final Project / Final Examination Instructions

May 8, 2026

1. Project Overview

The final examination for this course will be an **individual practical project** focused on applying deep learning techniques to solve a real-world problem in the biomedical, bioinformatics, or healthcare domain.

Students are expected to demonstrate:

- Understanding of deep learning concepts covered in class
- Ability to work with real-world biomedical or healthcare datasets
- Ability to design, train, evaluate, and interpret deep learning models
- Reproducible experimentation and scientific reporting
- Practical implementation and communication skills

The project should reflect the standards of a small-scale research or applied AI project similar to industry or academic work.

Possible application areas include:

- Medical imaging
- Bioinformatics
- Clinical NLP
- Computational biology
- Public health AI
- Genomics
- Microscopy AI
- Biomedical signal analysis
- Digital health systems

2. Important Dates

Activity	**Date**
Dataset submission for approval	**15 May**
Project presentations	**30 May starting at 9:00 AM**
Final report submission	**30 May**
GitHub repository submission	**30 May**

3. Project Rules and Constraints

1. This is an **individual project**.

2. Students must submit the **dataset link for approval by 15 May**. Failure to obtain dataset approval may result in a fail.

3. All work must be maintained in a **public GitHub repository**.

4. Projects must use datasets within the:

    - healthcare
    - biomedical
    - clinical
    - public health
    - genomics
    - bioinformatics
    - medical imaging

    domains.

5. Datasets from heavily explored platforms such as:

    - Kaggle
    - Zindi

    are discouraged unless strong justification is provided.

6. Simply reproducing existing notebooks or tutorials is not acceptable.

7. Students are encouraged to use:

    - transfer learning
    - explainability techniques
    - deployment tools
    - reproducible pipelines

8. Ethical considerations must be discussed where applicable, especially for:

    - patient data
    - fairness
    - privacy

- bias

9. Use of AI coding assistants is allowed for productivity purposes; however:
    - students must understand all submitted work
    - plagiarism is prohibited
    - copied solutions without understanding will receive penalties

4. Dataset Requirements

Datasets should:

- Be relevant to health, medicine, biology, or bioinformatics
- Preferably contain at least 1000 samples in raw form
- Have sufficient documentation or metadata
- Be legally usable for research or educational purposes

5. Example Project Areas

Project Area	Modality	Suggested Models	Difficulty
Malaria Detection	Microscopy Images	CNNs, ResNet	Medium
Chest X-ray Classification	Medical Imaging	DenseNet, EfficientNet	Medium
ECG Arrhythmia Detection	Time Series	CNNs, LSTMs	Medium
Tumor Segmentation	Medical Imaging	U-Net	High
Clinical NLP Classification	Text	Transformers, BERT	Medium
DNA Sequence Classification	Sequence Data	CNNs, Transformers	High
Drug Response Prediction	Tabular + Genomics	Deep Neural Networks	High
Maternal Risk Prediction	Tabular Data	MLPs, XGBoost	Medium
Antimicrobial Resistance Prediction	Genomics	Deep Learning	High
Air Quality and Health Modeling	Environmental Data	LSTMs, MLPs	Medium

6. Required Technical Components

Your project must include:

- Data exploration and visualization
- Data preprocessing

- Train/validation/test splitting
- Baseline model
- At least one deep learning model
- Appropriate evaluation metrics
- Error analysis
- Interpretation of results
- Discussion of limitations
- Reproducibility

Students are strongly encouraged to include:

- Transfer learning
- Explainability methods
- Deployment/demo interfaces
- Experiment tracking

7. Recommended Technical Stack

Suggested tools and frameworks include:

- PyTorch
- TensorFlow/Keras
- MONAI
- HuggingFace
- OpenCV
- scikit-learn
- Weights & Biases
- Streamlit
- Gradio
- CVAT
- Roboflow

8. Deliverables

Students must submit:

1. Approved dataset link
2. GitHub repository
3. Jupyter/Colab notebooks
4. Final report
5. Presentation slides
6. Optional deployment/demo

Expected GitHub Structure

```
project/

 data/
 notebooks/
 src/
 models/
 reports/
 figures/
 requirements.txt
 README.md
```

9. Final Report Structure

The report should be between **6–10 pages** and include:

1. Abstract
2. Introduction
3. Related Work
4. Dataset Description
5. Methodology
6. Experiments
7. Results
8. Error Analysis
9. Ethical Considerations
10. Limitations
11. Future Work
12. Conclusion
13. References

10. Presentation Guidelines

- Presentation duration: **10–15 minutes**
- Focus on:
  - problem motivation
  - methodology
  - results
  - lessons learned
- Do not overload slides with code.
- Demonstrations are encouraged.
- Presentations should tell a coherent technical story.

11. Evaluation Rubric

Component	**Marks**
Problem originality and relevance	10
Dataset quality and justification	10
Data preprocessing and exploration	10
Model implementation	15
Experimentation and evaluation	15
Results interpretation and analysis	10
Report quality	10
GitHub organization and reproducibility	10
Presentation quality	5
Deployment/demo (optional but rewarded)	5
Total	**100**

All the best with your projects.

The goal is not only to build accurate models, but also to build meaningful, reproducible, and impactful AI systems for health.